

a Language - set of strings

hello $\leftarrow$ spec.
yes
L

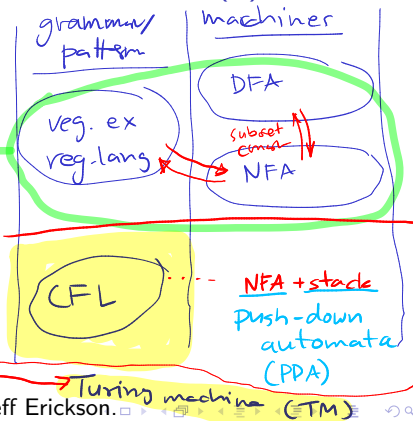
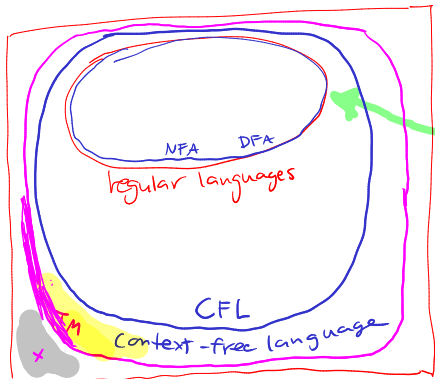
$$\{0^n 1^n \mid n \geq 0\}$$
$$\{0^n 1^n 0^n \mid n \geq 0\}$$

01204211 Discrete Mathematics

Lecture 11a: Context-free languages and grammars (1)¹

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undecidable
languages

High-level programming languages

- ▶ 1954 - FORTRAN

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- ▶ 1958 - LISP

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High-level programming languages

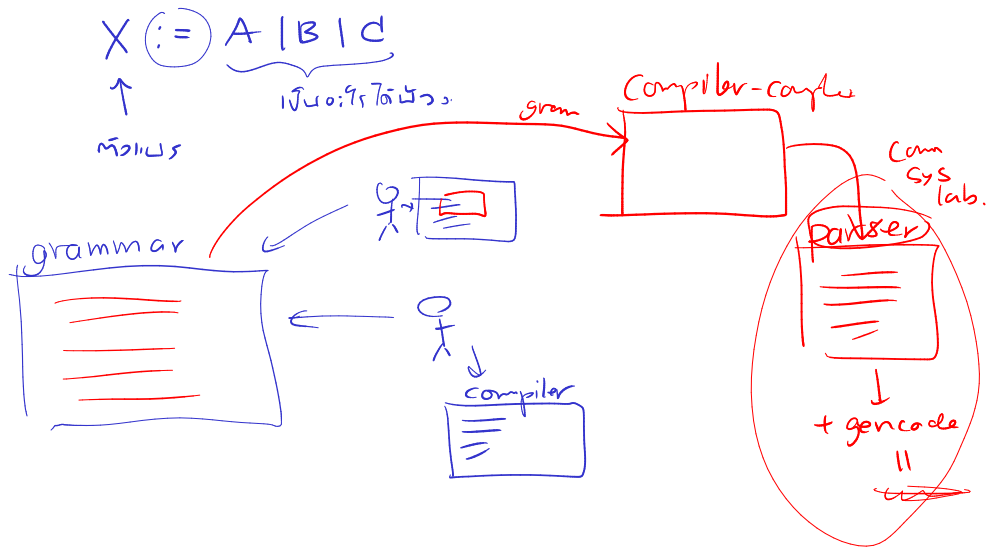
▶ 1954 - FORTRAN *x*

▶ 1958 - LISP

▶ 1958 - COBOL

▶ 1960 - ALGO60 *x* fun

Backus-Naur form



Python grammar

Building up languages

Regular languages:

- ▶ Contenation
- ▶ Union
- ▶ Kleene star

Building up languages

Regular languages:

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- ▶ Union
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Context-free grammar:

- ▶ Contenation
- ▶ Union
- ▶ Recursion

$$S \rightarrow \varepsilon \mid 1S0$$

generates

$$\{0^n 1^n \mid n \geq 0\} = \{\varepsilon, 1^0, 1100, 111000, \dots\}$$

$$1S0$$

$$1\varepsilon 0 = 10 \quad \checkmark$$

$$11S00$$

$$11\varepsilon 00 = 1100$$

Example

terminals
 $\Sigma = \{0, 1\}$

$\Gamma = \{S\}$

nonterminal s

$$\boxed{S} \rightarrow 0S1$$

$$S \rightarrow \varepsilon$$

$$S \rightarrow (0S1)$$

$$S \rightarrow (\varepsilon)$$

rule (n_g)

production rule.

$$\in (\Sigma \cup \Gamma)^*$$

Example

$$S \rightarrow 0S1$$

$$S \rightarrow \varepsilon$$

You can use “|” to write production rules more succinctly.

$$S \rightarrow 0S1 \mid \varepsilon$$

Definition

A **context-free grammar** consists of the following components:

- ▶ a finite set Σ , a set of *symbols* (or *terminals*),
- ▶ a finite set Γ disjoint from Σ , a set of *non-terminals* (you can think of them as *variables*),
- ▶ a finite set R of *production rules* of the form $A \rightarrow w$ where $A \in \Gamma$ and $w \in (\Sigma \cup \Gamma)^*$ is a string of symbols and variable, and
- ▶ a *starting non-terminal* (usually the non-terminal of the first production rule).

Another example

$$L(C) = \{0^n 1^n \mid n \geq 0\}$$

$$L(A) = \{0^m 1^n \mid m > n\}$$

$$L(B) = \{0^m 1^n \mid m < n\}$$

$$\begin{aligned} L(S) &= L(A) \cup L(B) \\ &= \{0^m 1^n \mid m \neq n\} \end{aligned}$$

Here $\Sigma = \{0, 1\}$ and $\Gamma = \{S, A, B, C\}$.

$$\begin{aligned} \underline{S} &\rightarrow A \mid B \\ A &\rightarrow 0A \mid 0C \\ B &\rightarrow B1 \mid C1 \\ C &\rightarrow \varepsilon \mid 0C1 \end{aligned}$$

For string w ,
 $w \in L(S)$

00011

S $\rightarrow$ A

$\rightarrow$ 0C

$\rightarrow$ 00C1

$\rightarrow$ 000C11

$\rightarrow$ 000 ε 11
 $= 00011$

derivation

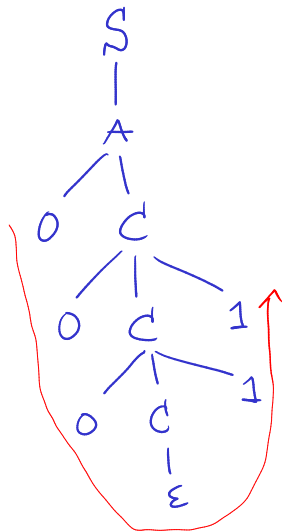
Another example

11100

$$\begin{aligned} S &\rightarrow A \mid B \\ A &\rightarrow 0A \mid 0C \\ B &\rightarrow B1 \mid C1 \\ C &\rightarrow \varepsilon \mid 0C1 \end{aligned}$$

Here $\Sigma = \{0, 1\}$ and $\Gamma = \{S, A, B, C\}$.

Parse tree.



Applying the rules

$$01\underline{\Sigma}10 \rightsquigarrow 01\underline{0S0}10$$

$$S \rightarrow 0S0$$

If you have strings $x, y, z \in (\Sigma \cup \Gamma)^*$ and the production rule

$$A \rightarrow y,$$

You can apply the rule to the string xAz . This yields the string

$$xyz.$$

We use the notation

$$xAz \rightsquigarrow xyz$$

to describe this application.

Derivation



We say that z derives from x if we can obtain z from x by production rule applications, denoted by $x \rightsquigarrow^* z$.

Formally, for any string $x, z \in (\Sigma \cup \Gamma)^*$, we say that x $\rightsquigarrow^*$ z if either

○ app' $\triangleright$ $x = z$, or

$\triangleright$ $x \rightsquigarrow y$ and $y \rightsquigarrow^* z$ for some string y $\in (\Sigma \cup \Gamma)^*$.



$L(w)$

The *language* $L(w)$ of string $w \in (\Sigma \cup \Gamma)^*$ is the set of all strings in Σ^* that derive from w , i.e.,

$$L(w) = \{x \in \Sigma^* \mid w \rightsquigarrow^* x\}.$$

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The language **generated by** a context-free grammar G , denoted by $L(G)$ is the language of its starting non-terminal.

A language L is **context-free** if there exists some context-free grammar G such that

$L(G) = L.$

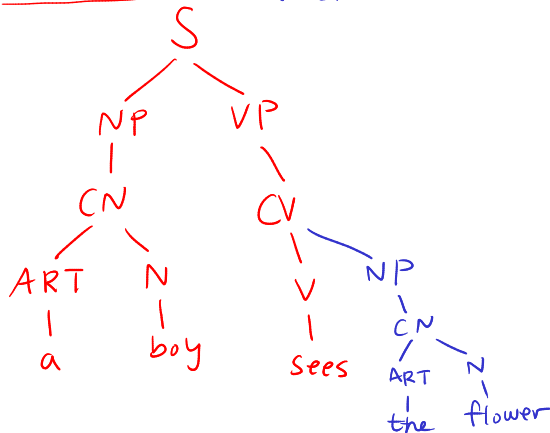
$S \rightarrow NP VP$
 $NP \rightarrow CN | CN PP$
 $VP \rightarrow CV | CV PP$
 $PP \rightarrow PREP CN$
 $CN \rightarrow ART N$
 $CV \rightarrow V | V NP$
 $ART \rightarrow a | the$
 $N \rightarrow boy | girl | flower$
 $V \rightarrow touches | likes | sees$
 $PREP \rightarrow with$

Small English grammar

$S \rightarrow NP VP$
 $NP \rightarrow \underline{CN} | CN PP$
 $VP \rightarrow \underline{CV} | CV PP$
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► Examples of strings in $L(G_2)$ are:

► a boy sees the flower



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► Examples of strings in $L(G_2)$ are:

- a boy sees
- the boy sees a flower
- a girl with a flower likes the boy

Parse tree

$$S \rightarrow A \mid B$$

$$A \rightarrow 0A \mid 0C$$

$$B \rightarrow B1 \mid C1$$

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Parse tree

► 00011

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Parse tree

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► 01111

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Parse tree

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► 111110

Parse tree

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Parse tree

a girl with a flower likes the boy

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$VP \rightarrow CV|CV PP$

$PP \rightarrow PREP CN$

$CN \rightarrow ART N$

$CV \rightarrow V|V NP$

$ART \rightarrow a|the$

$N \rightarrow boy|girl|flower$

$V \rightarrow touches|likes|sees$

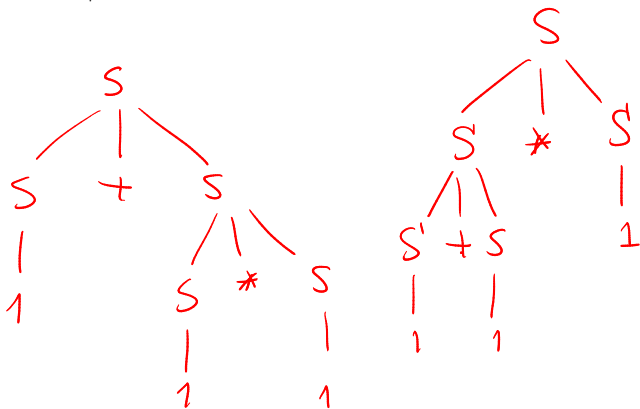
$PREP \rightarrow with$

Ambiguity

$a+b*c$

► $1 + 1 * 1$

$S \rightarrow 1 \mid S + S \mid S * S$



Ambiguity

► $1 + 1 * 1$

► $1 + 1 + 1 + 1 + 1$

$$S \rightarrow 1 \mid S + S \mid S * S$$

Ambiguity

▶ $1 + 1 * 1$

▶ $1 + 1 + 1 + 1 + 1$

$$S \rightarrow 1 \mid S + S \mid S * S$$

- ▶ A string w is **ambiguous** with respect to a grammar G if more than one parse tree for w exists.
- ▶ A grammar G is **ambiguous** if some string is ambiguous with respect to G .